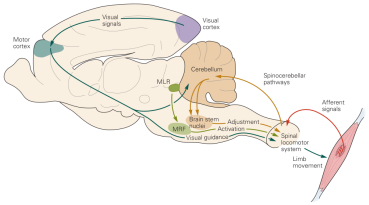
PHYSIOLOGY OF THE MOTOR SYSTEM

Principles of Sensorimotor Control

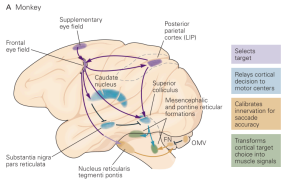
Lecture 01

Different motor tasks, different circuit layouts

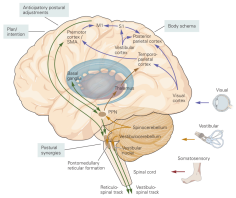
LOCOMOTION



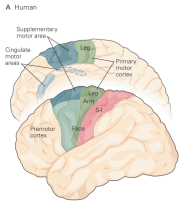
GAZE



POSTURE



VOLUNTARY ACTION



> These tasks draw on differently organized, interconnected circuits.

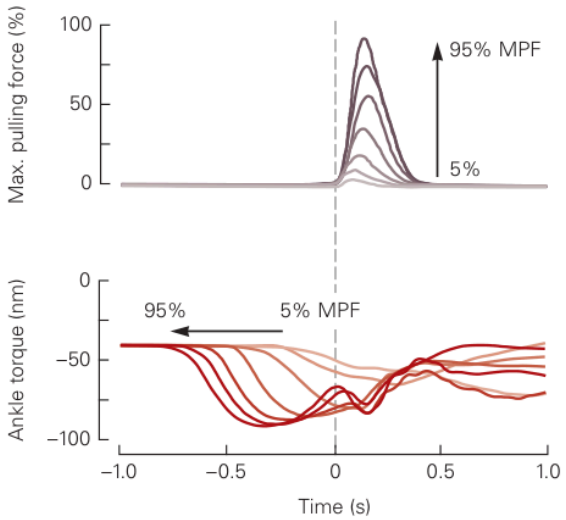
Kandel 6e, Figs. 33–1, 35–9A, 36–13, 34–2A · selective circuit maps.

LECTURE NOTE

The four circuit maps show that different motor tasks recruit differently organized, interconnected pathways. This opens the lecture's central question: what control principles can these partly distinct systems share?

YOUR NOTES

Does ankle torque change before the arm pull?



Kandel 6e, Fig. 36–8A, p. 893 · arm force and ankle torque

READ THE TRACES

Time zero: pulling force begins at the handle. When does ankle torque change?

> Ankle torque changes before the measured arm pull.

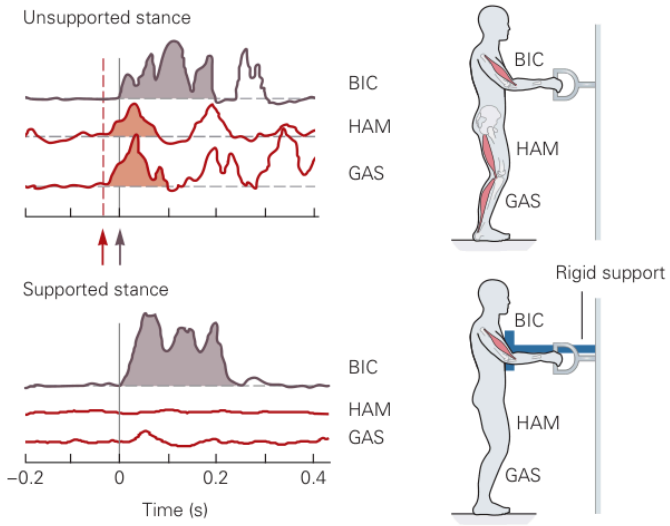
A.02 2 / 38

LECTURE NOTE

Ankle torque changes before force appears at the handle, suggesting that pulling includes preparation for its postural consequences. This first trace shows motor systems cooperating within one action; the support manipulation next helps establish why the leg response occurs.

YOUR NOTES

Does external support change postural recruitment?



Kandel 6e, Fig. 36–8B, p. 893 · EMG and support manipulation

COMPARE CONDITIONS

Time zero: biceps activation. Compare leg muscle traces with and without rigid support.

> Hamstring and gastrocnemius recruitment is reduced with support.

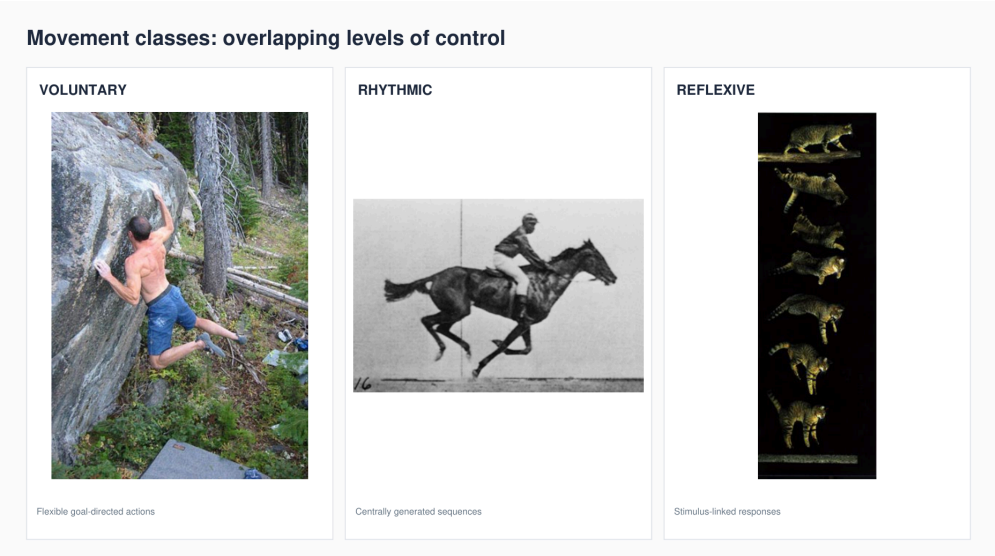
A.03 3 / 38

LECTURE NOTE

External support reduces leg-muscle recruitment while the arm still pulls, strengthening the case that the earlier leg response serves postural demands. Together, the timing and support results show why task-specific motor systems must coordinate rather than operate independently.

YOUR NOTES

Voluntary, rhythmic, and reflexive are control modes



Kandel 6e, Ch. 30, pp. 715–716; adapted prior L01 examples · conceptual illustration

LECTURE NOTE

Voluntary, rhythmic, and reflexive describe ways movements are controlled, not separate sets of body parts. After distinguishing motor systems by their tasks, we now use these modes to compare the kinds of control each task may require.

YOUR NOTES

The modes differ in typical control demands

VOLUNTARY	RHYTHMIC	REFLEXIVE
Flexible, goal-dependent initiation	Recurrent patterns with ongoing control	Relatively rapid, stimulus-linked response

> The same action may combine all three modes.

Kandel 6e, Ch. 30, pp. 715–716 · teaching synthesis.

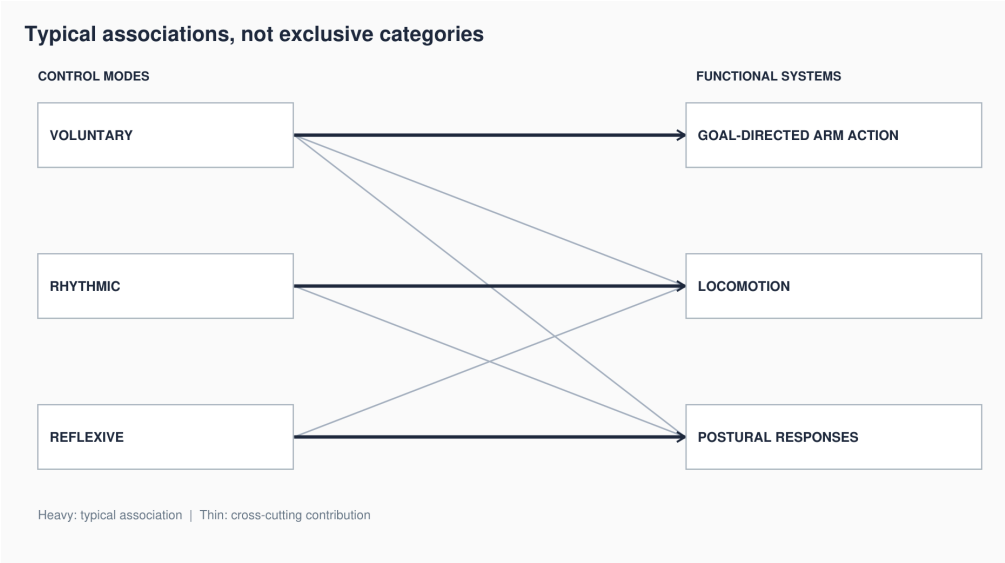
B.02 5 / 38

LECTURE NOTE

The three modes differ in their usual demands for goal selection, repetition, and rapid response, yet one action can combine them. This keeps the mode framework flexible enough to describe the same functional systems we met at the start.

YOUR NOTES

Modes cross the functional systems



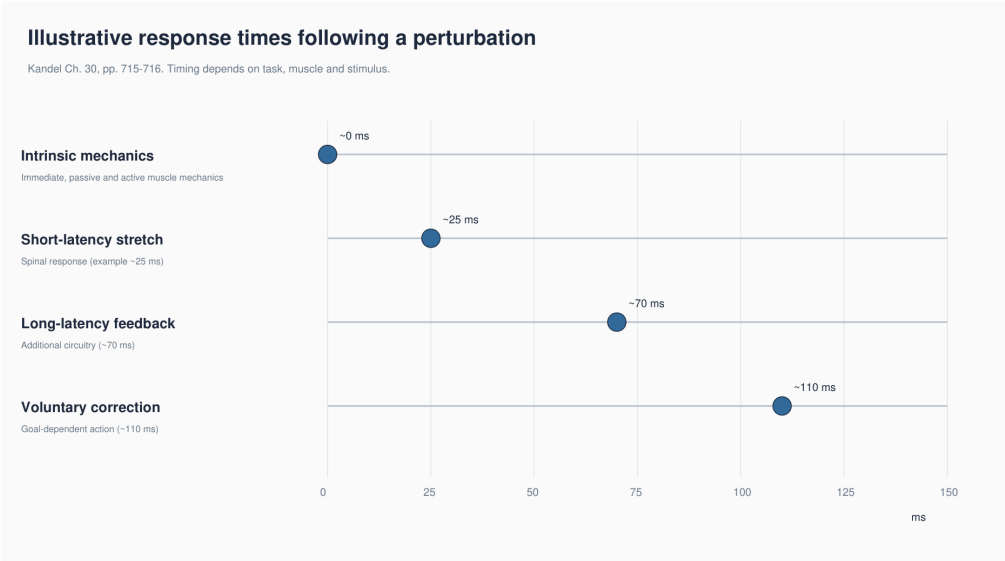
Kandel 6e, Ch. 30 and systems chapters · instructor–specialist conceptual map

LECTURE NOTE

Gaze and walking each recruit more than one control mode, so a functional system cannot be assigned a single place in the mode scheme. With tasks and modes distinguished, we can ask how mechanics and neural responses cooperate over time.

YOUR NOTES

Mechanics and neural responses occupy different timescales



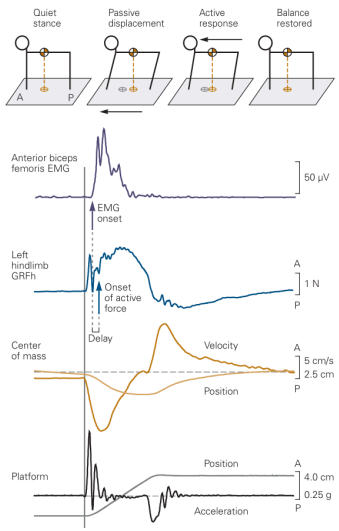
Kandel 6e, Ch. 30, pp. 715–716 · illustrative latencies

LECTURE NOTE

A perturbation first acts through body mechanics; newly evoked neural responses follow on different, overlapping timescales. This timing view helps us read the postural recording next as a sequence of interacting events rather than a choice between reflexive and voluntary control.

YOUR NOTES

A postural perturbation produces several measured responses



Kandel 6e, Fig. 36–3, p. 888 · EMG and force recordings

READ DOWN THE TRACES

Platform motion, muscle activity, center of mass, and corrective motion unfold over time.

> The measured response is a coupled mechanical and neural process.

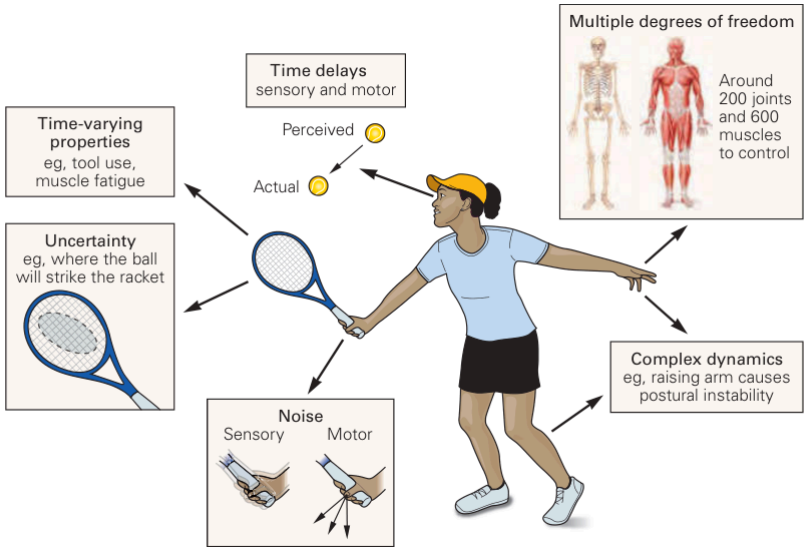
B.05 8 / 38

LECTURE NOTE

Platform movement shifts the body; muscle activity then precedes active force and recovery of body position. This sequence shows why postural control must cope with immediate mechanics and delayed neural action, the shared challenge we examine next.

YOUR NOTES

One movement faces several control challenges



Kandel 6e, Fig. 30–1, p. 714 · schematic illustration

C.01 9 / 38

LECTURE NOTE

Even a simple action must manage coupled joints, uncertain dynamics, many possible muscle patterns, and delayed sensation. Having seen these demands in particular motor tasks, we now name the general problems that the rest of the lecture's strategies will address.

YOUR NOTES

Many joints and muscles serve a few goals

BEHAVIORAL GOAL

Place the hand at the target

CONTROL VARIABLES

Coordinate joints and muscles whose effects are coupled

> The controller must manage many possible solutions under uncertain, delayed feedback.

Kandel 6e, Fig. 30–1, pp. 714–715 · conceptual synthesis.

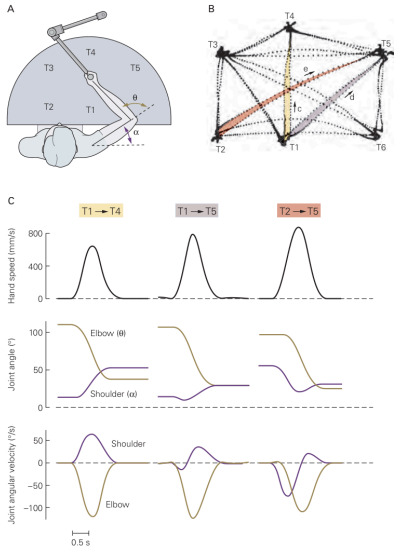
C.02 10 / 38

LECTURE NOTE

Reaching one hand target leaves many possible joint and muscle solutions, so specifying the goal does not specify the commands. This degrees-of-freedom problem gives the abstract challenge diagram a concrete form before we examine actual movement paths.

YOUR NOTES

Similar hand paths can arise from complex joint motion



Kandel 6e, Fig. 30–8A–C, p. 726 · reaching kinematics

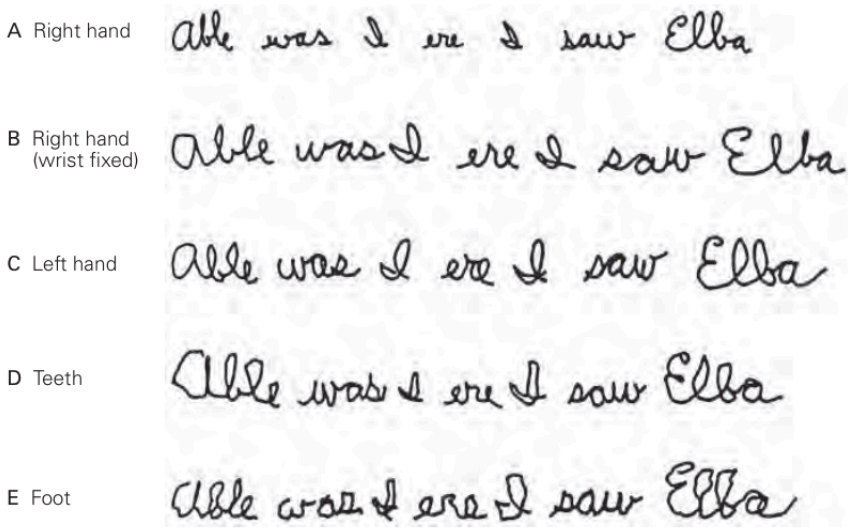
C.03 11 / 38

LECTURE NOTE

A simple hand path can arise from intricate, coupled joint motion, showing how much coordination lies beneath one behavioral goal. This links the many-solutions problem to measured movement and prepares a second example in which the same goal survives a change of muscles.

YOUR NOTES

A recognizable letter survives a change of effector



Kandel 6e, Fig. 30–9A–E, p. 727 · handwriting observation

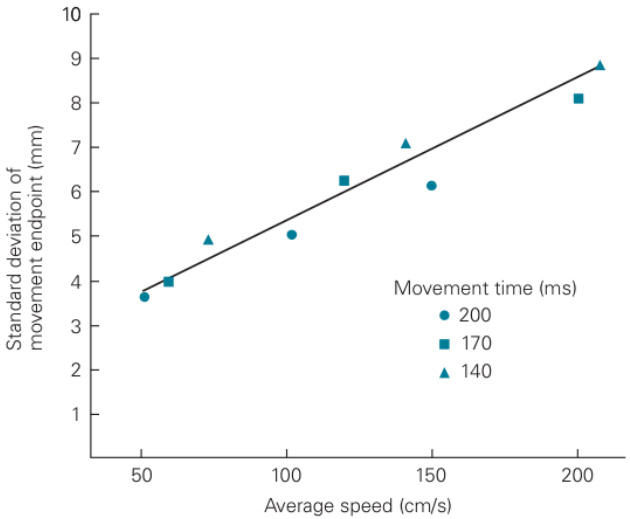
C.04 12 / 38

LECTURE NOTE

A recognizable letter survives a change from hand to teeth or foot, showing that the task can be achieved through different muscle patterns. Together with the reaching paths, this separates the behavioral goal from any single joint or muscle solution.

YOUR NOTES

Faster reaches show more endpoint variability



Kandel 6e, Fig. 30–10, p. 728 · endpoint SD vs speed

READ THE Y AXIS

This is variability, not mean directional bias.

> Speed and precision trade off in these data.

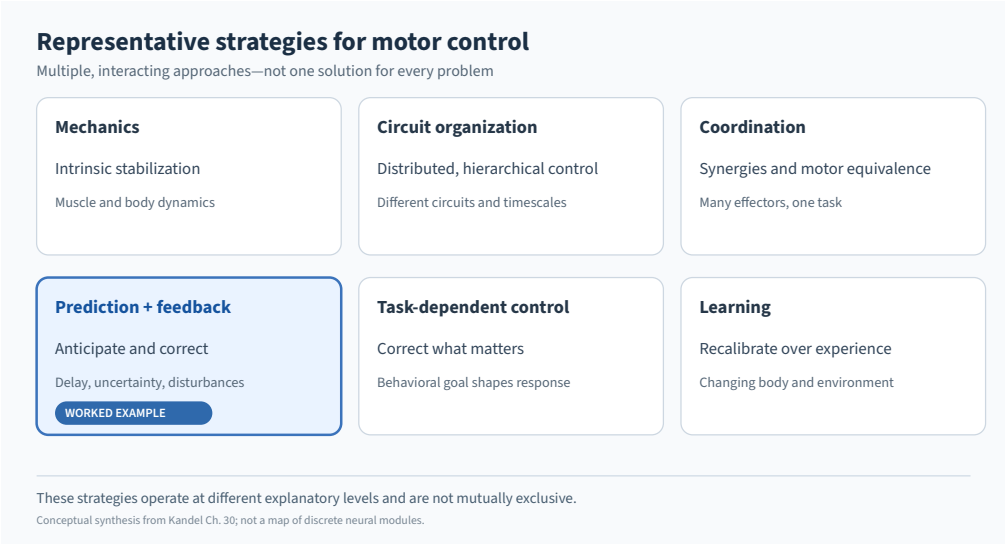
C.05 13 / 38

LECTURE NOTE

Endpoint spread rises with reach speed, so choosing among possible movements also involves a precision cost. This adds uncertainty to the degrees-of-freedom problem and motivates the complementary strategies we turn to next.

YOUR NOTES

Motor systems use complementary strategies



Kandel 6e, Ch. 30 · conceptual synthesis, not separate brain modules

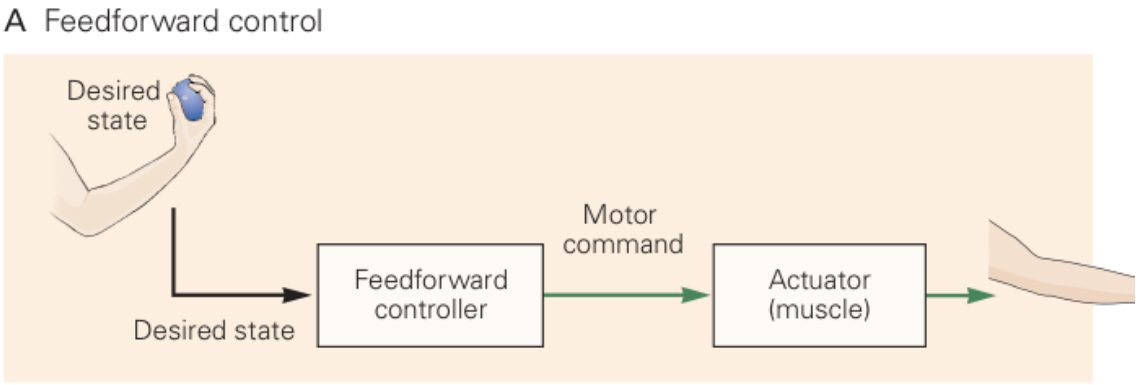
D.01 14 / 38

LECTURE NOTE

Mechanics can act immediately, prediction prepares for expected effects, feedback corrects mismatches, and learning changes later commands. These answers to delay and uncertainty guide the worked prediction-and-feedback examples that follow.

YOUR NOTES

Feedforward commands act before new sensory correction



Kandel 6e, Fig. 30–2A, p. 717 · model diagram

> Advance control can be fast, but an unforeseen disturbance is not corrected by this pathway alone.

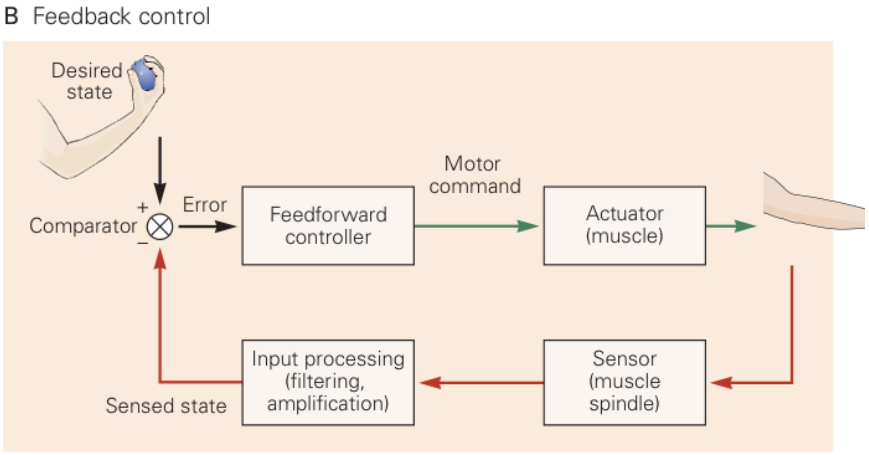
D.02 15 / 38

LECTURE NOTE

An advance command can act before new sensory consequences return, making it useful when movement is fast or feedback is delayed. It addresses the timing problem from the previous section, while leaving unexpected disturbances for feedback to handle.

YOUR NOTES

Feedback compares an outcome with a desired state



Kandel 6e, Fig. 30–2B, p. 717 · model diagram

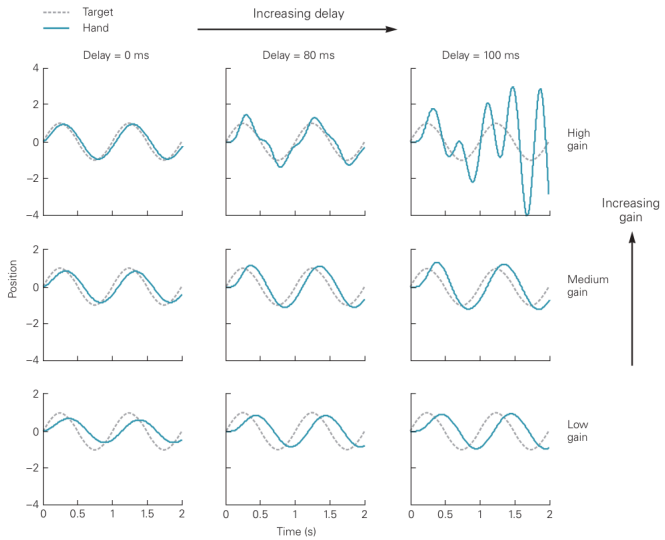
> Online correction can address disturbances, but its sensory estimate arrives with delay.

LECTURE NOTE

Feedback can adjust a command when a sensed result differs from the goal, addressing disturbances that advance control cannot foresee. Paired with the previous slide, it also raises a new question: what happens when the correction itself arrives late?

YOUR NOTES

Gain and delay jointly determine tracking



Kandel 6e, Fig. 30–4, p. 720 · controller simulation

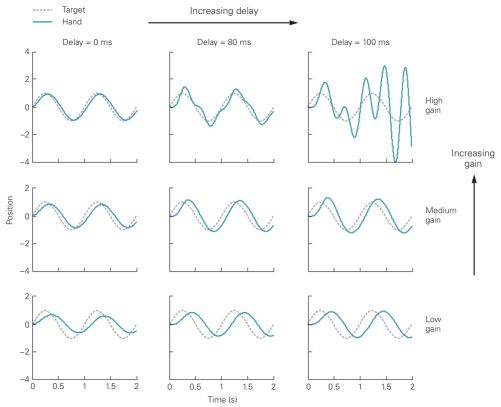
LECTURE NOTE

Across a row, more delay can make correction late; up a column, greater gain can produce oscillation when delay is sufficient. This simulation turns the feedback timing problem into a stability question that the next slide distills.

YOUR NOTES

Strong delayed correction can oscillate

More correction is not always better when the signal describes an earlier state.



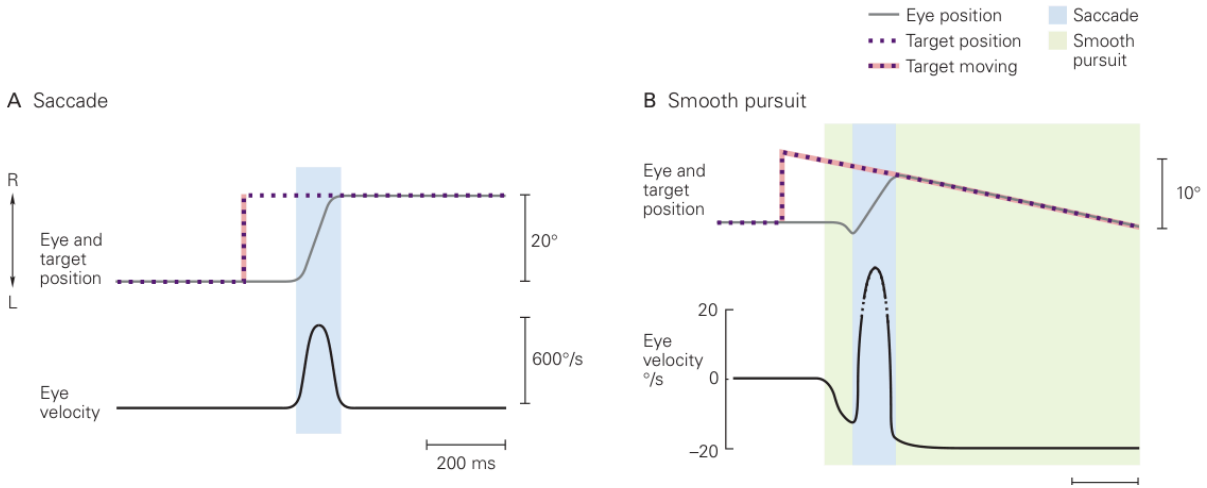
Kandel 6e, Fig. 30–4, p. 720 · simulation, not neural recording

LECTURE NOTE

A strong response to an old error may arrive after the body has moved, causing overshoot instead of stability. This summarizes the gain-and-delay example and explains why motor systems may need prediction alongside sensory correction.

YOUR NOTES

Gaze uses control strategies on different timescales



Kandel 6e, Fig. 35–6A/B, p. 867 · eye movement recordings

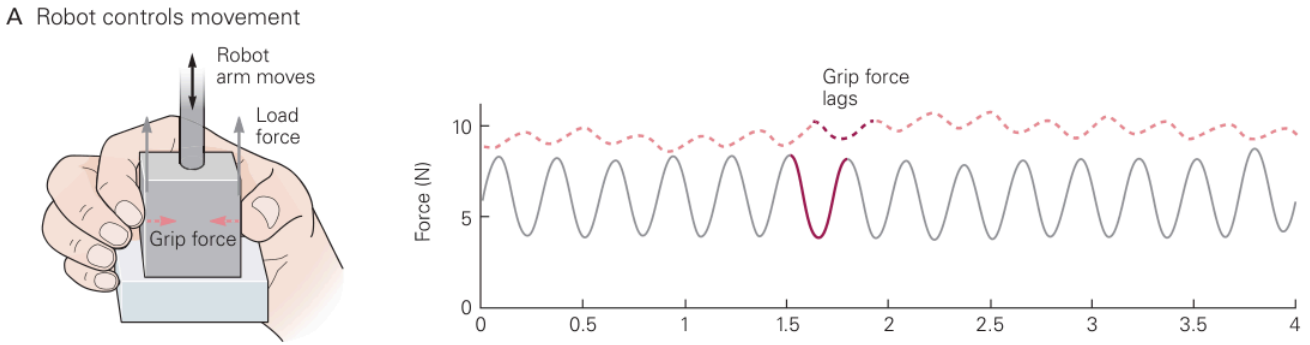
D.06 19 / 38

LECTURE NOTE

Fast saccades leave little time for fresh retinal correction, whereas pursuit can use ongoing visual motion. Gaze thus gives a real motor-system example of the timing tradeoff behind advance and feedback control before we turn to grip.

YOUR NOTES

External loading elicits a delayed grip response



Kandel 6e, Fig. 30–6A, p. 724 · externally produced load

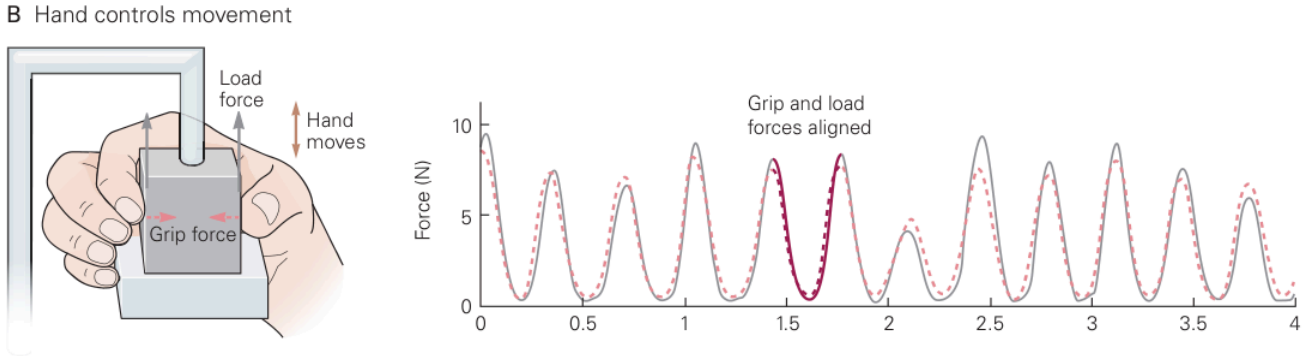
PINK DASHED · GRIP FORCE | GRAY SOLID · LOAD FORCE

LECTURE NOTE

When an external load changes, grip adjustment follows rather than matching it in time, consistent with reliance on sensed load. This provides the comparison needed to recognize anticipatory grip control when the person generates the load.

YOUR NOTES

Self-generated loading aligns grip and load force



Kandel 6e, Fig. 30–6B, p. 724 · self-produced load

PINK DASHED · GRIP FORCE | GRAY SOLID · LOAD FORCE

> The closer phase match supports anticipatory control.

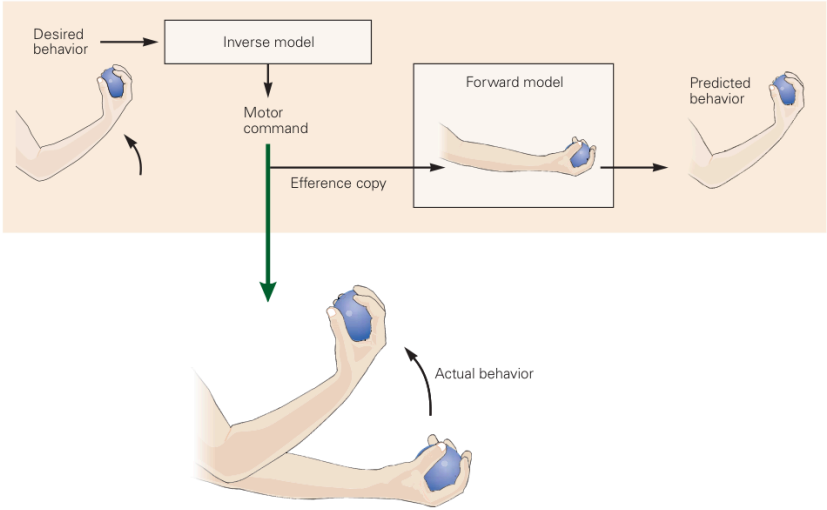
D.08 21 / 38

LECTURE NOTE

With a self-produced load, grip force rises closer in time to load force, supporting anticipation of the command's consequences. Compared with the preceding external-load trace, this is behavioral evidence for prediction rather than correction alone.

YOUR NOTES

Internal models propose distinct computations



Kandel 6e, Fig. 30–3, p. 718 · forward and inverse model schematic

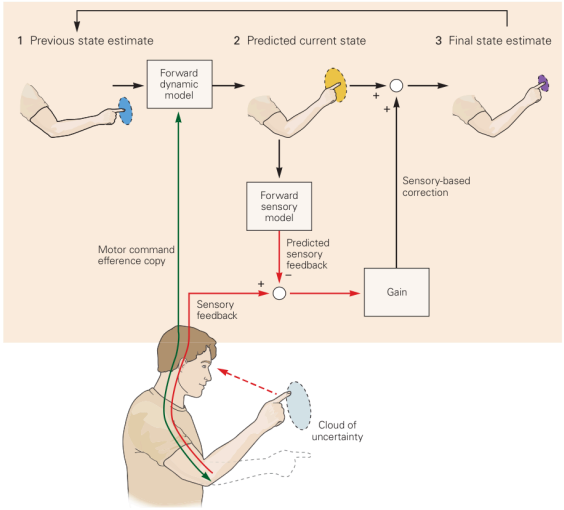
D.09 22 / 38

LECTURE NOTE

A forward model predicts the consequences of a command; an inverse model selects a command for a desired outcome. After the grip comparison revealed anticipatory adjustment, these computations offer a framework for how advance control might be organized.

YOUR NOTES

An observer combines prediction with sensation



Kandel 6e, Fig. 30–5, p. 722 · state-observer model

D.10 23 / 38

LECTURE NOTE

An observer model updates an estimated body state from the command and then corrects it with incoming sensation. It joins the predictive computation just introduced to the delayed feedback problem, offering a way to act on an estimate of the present.

YOUR NOTES

State estimation addresses delayed, noisy feedback

PREDICTION

Use the previous estimate and a copy of the command

MEASUREMENT

Use incoming sensory evidence to correct the estimate

> Neither source alone is sufficient across all conditions.

Kandel 6e, Figs. 30–3 and 30–5, pp. 718, 722 · model synthesis.

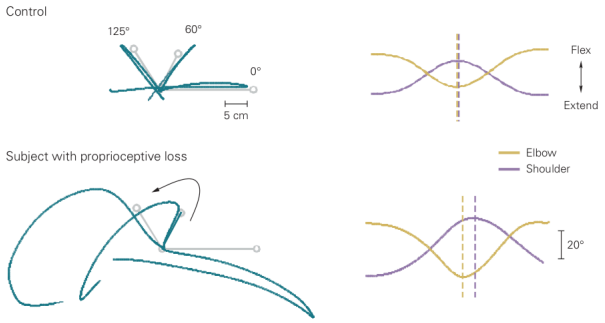
D.11 24 / 38

LECTURE NOTE

Prediction can bridge sensory delay, while incoming sensation can correct a prediction that drifts. This is the central synthesis of the model sequence; we next ask what happens to coordinated movement when usable sensory state information is lost.

YOUR NOTES

Loss of proprioception disrupts rapid reversal



Kandel 6e, Fig. 30–14, p. 733 · control and deafferented patient

COMPARE THE TRACES

How do limb paths and joint timing differ at the reversal?

> Usable state information matters for multijoint coordination.

D.12 25 / 38

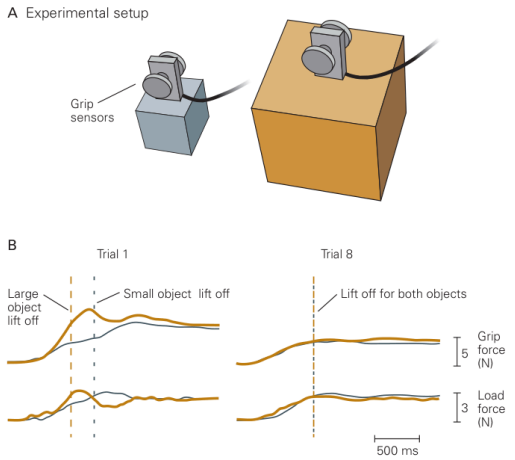
LECTURE NOTE

Without proprioception, the patient's paths and joint timing change markedly at rapid reversal, showing the value of usable state information for multijoint control. The comparison gives behavioral weight to the observer discussion without identifying its neural implementation.

YOUR NOTES

Grip scaling changes while heaviness perception persists

EQUAL WEIGHT · ORANGE = LARGE · GRAY = SMALL



Kandel 6e, Fig. 30–7A/B, p. 725 · size–weight illusion experiment

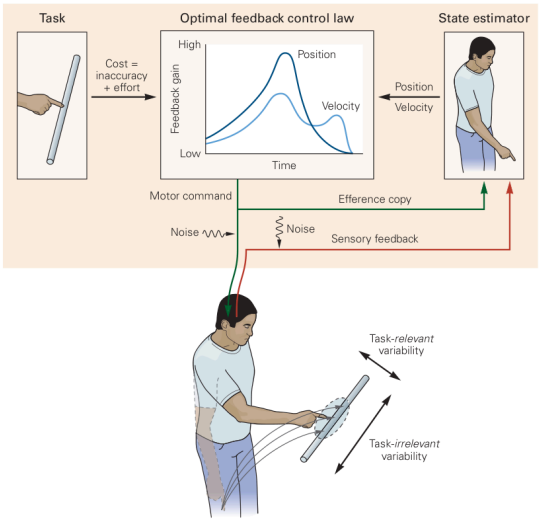
D.13 26 / 38

LECTURE NOTE

Grip and load forces adjust to equal-weight objects even while the smaller object continues to feel heavier. This extends the prediction-and-feedback discussion by showing that motor calibration and conscious judgment can change on different time courses.

YOUR NOTES

Task goals determine which deviations need correction



Kandel 6e, Fig. 30–11, p. 729 · optimal feedback control model

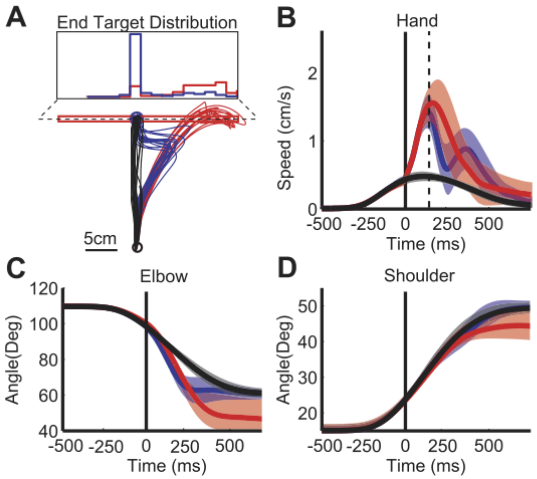
LECTURE NOTE

A displacement along a bar may still satisfy the goal, whereas the same displacement can miss a dot. After asking how the body estimates its state, we now ask which estimated errors matter enough to correct.

YOUR NOTES

A dot and a bar set different correction demands

BLUE · DOT TARGET | RED · BAR TARGET



Nashed et al. 2012, Fig. 2A–D, p. 1003 · perturbation kinematics

D.15 28 / 38

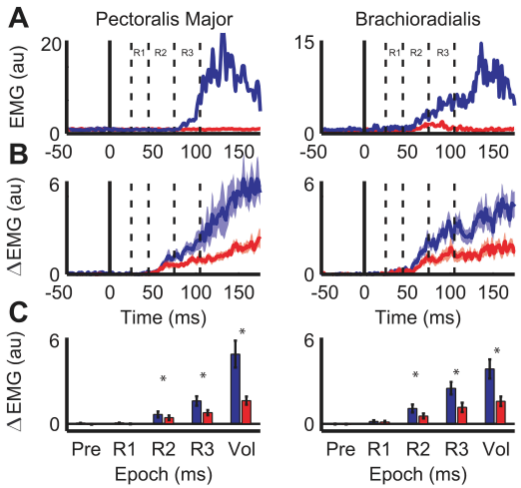
LECTURE NOTE

The same hand perturbation prompts a correction toward the dot but little path correction along the bar, because success depends on target geometry. This behavioral result supports the preceding goal-based model and sets up a test of the muscle response.

YOUR NOTES

Later muscle responses reflect task relevance

BLUE · DOT TARGET | RED · BAR TARGET



Nashed et al. 2012, Fig. 3A–C, p. 1003 · EMG

COMPARE EPOCHS

Is the early R1 response or the later R2/R3 response more task sensitive?

> Later perturbation responses differ with target geometry.

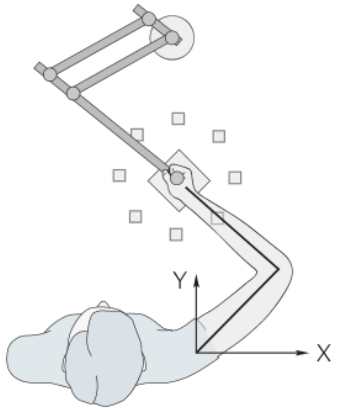
D.16 29 / 38

LECTURE NOTE

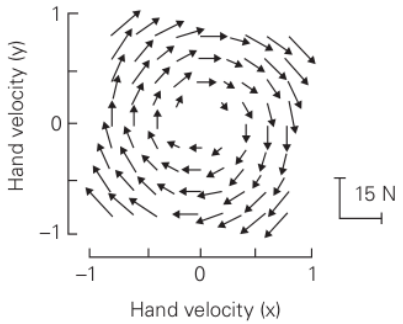
The early muscle response is similar across targets, but later activity differs with the dot or bar goal. This adds physiological evidence to the hand-path comparison: feedback can become sensitive to the action's purpose, not just the disturbance.

YOUR NOTES

A robotic handle applies a velocity-dependent force



Kandel 6e, Fig. 30–12A, p. 731 · apparatus



Kandel 6e, Fig. 30–12C, p. 731 · applied force field

> The device perturbs the hand during reaches to different targets.

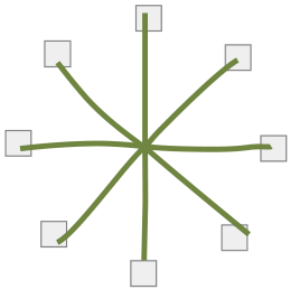
LECTURE NOTE

The robot changes the forces acting on the hand while the person reaches, creating a known disturbance whose effects can be followed across trials. This begins our learning section by asking whether control changes with experience, not only within a movement.

YOUR NOTES

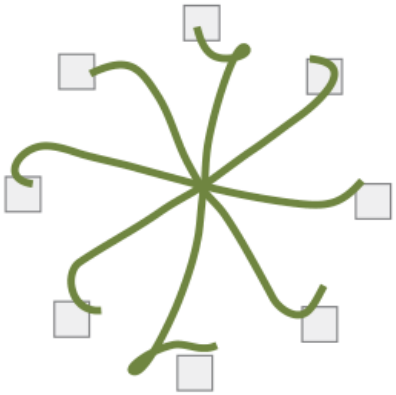
What changes when the force field turns on?

NULL FIELD



Kandel 6e, Fig. 30–12B, p. 731 · null-field paths

FIELD ON · FIRST REACHES



Kandel 6e, Fig. 30–12D1, p. 731 · initial paths

> With the force field on, initial paths curve away from the null-field paths.

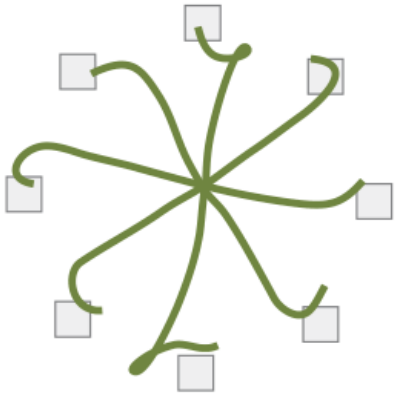
LECTURE NOTE

The first reaches curve when the field turns on, revealing a mismatch between current commands and the new dynamics. This initial error establishes the reference against which practice and later field removal can reveal adaptation.

YOUR NOTES

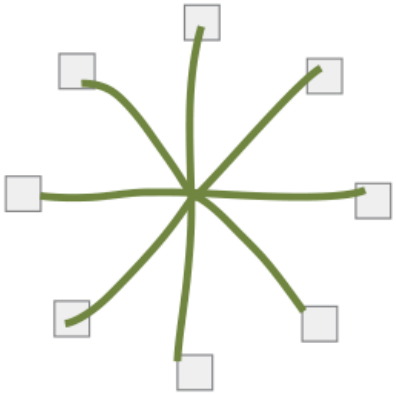
What changes with practice in the field?

FIELD ON · FIRST REACHES



Kandel 6e, Fig. 30–12D1, p. 731 · first reaches

FIELD ON · AFTER PRACTICE



Kandel 6e, Fig. 30–12D2, p. 731 · later reaches

> Path improvement while the perturbing field remains supports compensation.

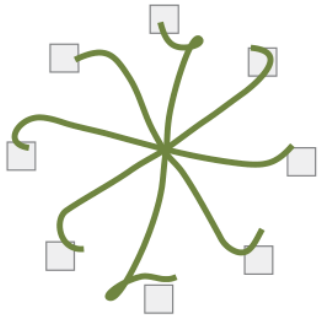
LECTURE NOTE

Hand paths straighten with practice even though the field remains, showing compensation for the altered dynamics. Improvement during exposure is only part of the learning case; removing the field next tests whether the changed control persists.

YOUR NOTES

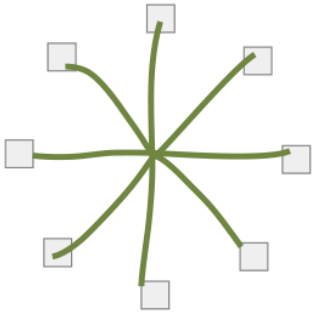
What happens when the field is removed?

FIELD ON · FIRST



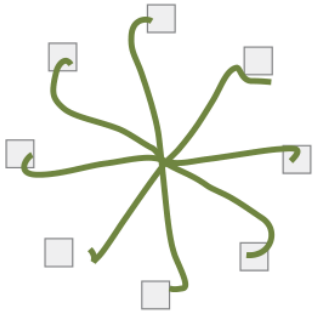
Kandel 6e, Fig. 30–12D1, p. 731

FIELD ON · LATER



Kandel 6e, Fig. 30–12D2, p. 731

FIELD OFF



Kandel 6e, Fig. 30–12D3, p. 731

> The deviation after removal is opposite the initial one: an aftereffect of learned compensation.

LECTURE NOTE

The field-off path bends opposite the initial field-on error, as though compensation continues after it is no longer needed. This aftereffect strengthens the learning inference and raises the question of which errors drove that change.

YOUR NOTES

Prediction error and target error answer different questions

PREDICTION ERROR

Was the sensed consequence different from the expected one?

TARGET ERROR

Did the action miss its behavioral goal?

> A predicted miss can have target error with little surprise.

Kandel 6e, pp. 718, 730–731; Figs. 30–3 and 30–12 · conceptual contrast.

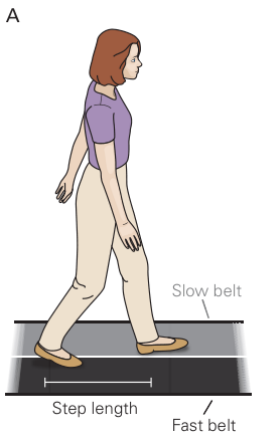
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LECTURE NOTE

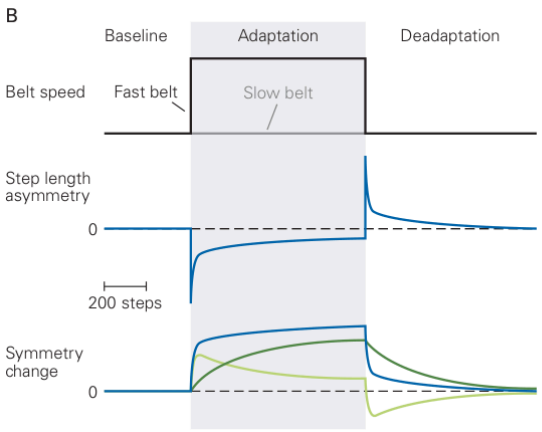
A movement can reach its goal yet have an unexpected sensory consequence, or miss as expected, so surprise and target failure need not coincide.
Distinguishing these errors helps interpret the force-field result before we look for adaptation in walking.

YOUR NOTES

Split-belt walking changes step symmetry



Kandel 6e, Fig. 30–13A, p. 732 · treadmill



Kandel 6e, Fig. 30–13B, p. 732 · illustrated time course and model

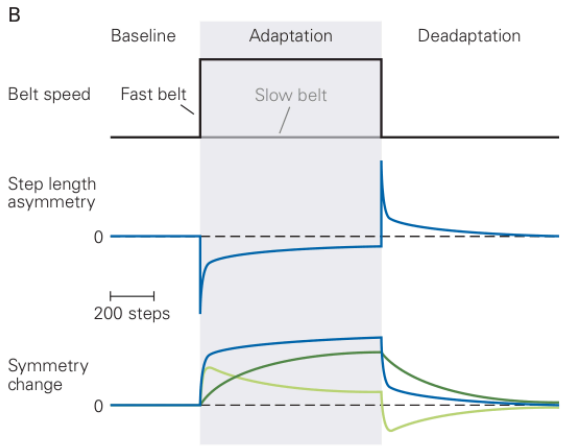
> Step asymmetry appears, diminishes with practice, and reverses transiently at washout.

LECTURE NOTE

Step asymmetry decreases during split-belt practice and briefly reverses when equal belt speeds return, an aftereffect like the one seen in reaching. This carries the learning argument into gait while preserving the differences between the two tasks.

YOUR NOTES

Fast and slow processes can fit the gait time course



Kandel 6e, Fig. 30–13B, p. 732 · two-rate model fit

> The curves summarize different adaptation rates; they are not recordings of two neuron populations.

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LECTURE NOTE

In the model, the fast component responds and fades quickly, while the slow component builds and persists longer; their sum follows the observed gait course. This puts the walking aftereffect into a learning framework without treating model components as identified circuits.

YOUR NOTES

Skill learning poses a wider credit-assignment problem

ADAPTATION	SKILL ACQUISITION
Compensate for an identifiable perturbation	Improve a policy when many choices contribute to delayed success

> Adaptation and skill learning overlap, but solve different parts of motor learning.

Kandel 6e, Ch. 30, pp. 732–735 · reading synthesis.

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LECTURE NOTE

Force-field adaptation has a defined disturbance and fairly immediate error, whereas skill improvement can depend on many choices and delayed outcomes. This broadens the lecture's learning account from compensation to deciding which actions deserve credit.

YOUR NOTES

Common principles, different implementations

GAZE

Rapid orientation and stabilization

STANCE

Stability amid perturbation

GAIT

Rhythm, guidance, and adaptation

REACHING

Flexible, task-dependent goals

> Prediction, feedback, and learning recur across systems with partly distinct circuitry.

Kandel 6e, Ch. 30 and Figs. 30–1, 30–11, 30–13 · lecture synthesis.

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LECTURE NOTE

The grip comparison, target-dependent correction, and aftereffects exposed three recurring demands: anticipate consequences, correct what matters, and change future commands. Together they explain how the systems introduced at the start can share control principles while relying on partly distinct circuits.

YOUR NOTES